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IST3134 BIG DATA ANALYTICS IN THE CLOUD

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1.0 Introduction

1.1 Background

Business performance and customer behaviour can be understood by using customer-generated data that are generated in large columns by the rapid growth of online review platforms. One of the examples of these online platforms is Yelp where it contains information about businesses, customer reviews, ratings, and user interactions. This information provides a valuable source of data to analyse how businesses perform from multiple standpoints. However, customer ratings alone are not enough to represent the business performance accurately. A business with a high rating but has a small number of reviews may contain less evidence of sustained satisfaction of customers compared to businesses with lower ratings but have thousands of reviews and substantial engagement by customers.

Yelp Open Dataset was used in this study to analyse how businesses perform by combining customer satisfaction, customer engagement, and volume of reviews. By integrating Business, Review and User datasets from the Yelp Open Dataset, analysis was done to provide a broader view of the business activity. There are approximately 6.99 million review records in the integrated dataset. This dataset is large enough to demonstrate the techniques of the application of big data processes.

To calculate the Business Success Score, the Business Success Analysis algorithm was developed to qualify businesses. Weighted components were used to combine average review rating, review volume, and customer engagement to obtain the score. Broader patterns in business performance were identified by extending the analysis to business categories, cities and states.

Apache Spark by using PySpark was used as the primary implementation to perform the data cleaning, integration, aggregation and feature engineering on the large dataset. Additionally, a non big data approach which was a traditional Pandas-based implementation was used to perform a similar analysis. By comparing the two approaches, it provided an opportunity in this study to examine how a distributed big data technology such as Spark can differ from conventional approach when applied to a large dataset.

1.2 Problem

Although traditional businesses often heavily rely on the customer ratings to be ranked, ratings alone are insufficient to provide a comprehensive representation of business performance. For example, businesses with a small number of highly positive reviews can receive high ratings whereas businesses with a high number of reviews can receive consistently strong ratings. Similarly, the Useful, Funny and Cool votes provide additional information on customer engagement about how the reviews are being interacted by the customers. However, they may be overlooked when ratings alone are used to rank the businesses.

The size of the Yelp dataset also posed another challenge as the Review datasets contained millions of records which made the integration of data, aggregation, and analysis even more computationally demanding compared to when working with a small conventional dataset. An efficient approach is needed to process such data so operations such as joining multiple datasets, aggregating millions of reviews and generating metrics on business-level can be performed.

As a result, this study addresses the problem of how to identify and compare successful Yelp businesses using multiple performance indicators and at the same time, process a large-scale review dataset efficiently. Business Success Score is obtained by proposing an approach that combines customer satisfaction, popularity and customer engagement. The big data processing approach used was Apache Spark whereas the traditional approach used for comparison was Pandas.

The analysis considered the reliability of the resulting rankings as well. This is due to businesses with very few records that may produce ratings that are unstable. In the meantime, misleading aggregate results may also be produced due to the small number of businesses representing cities or categories. Thus, to improve the reliability and the fairness of the comparisons to avoid bias, minimum thresholds and sensitivity analysis was incorporated into the methodology.

1.3 Research Question

The main research question of this study is how Apache Spark can be utilized efficiently to analyze large-scale review data by Yelp and identify businesses that are successful based on customer satisfaction, review volume, and customer engagement.

2.0 Dataset description and integration

2.1 Dataset Description

The Yelp Open Dataset was used in this study. This dataset is a publicly available dataset that has a collection of real-world business and customer review data provided by Yelp. Millions of records on businesses, user reviews, user profiles, business attributes, operating hours and customer interactions are contained in this dataset. This dataset is widely used in academic research and big data analytics for its large size, realistic structure and the interconnected relationships between multiple datasets.

The dataset was downloaded from the official website of the Yelp Open Dataset:

<https://business.yelp.com/data/resources/open-dataset/>

Three JSON datasets were chosen because they contained all the information needed to evaluate the business success from the perspectives of both business and customer.

Dataset	Records	Purpose
Business	150,346	Stores information on each business including the name, location, categories, average Yelp rating, review numbers, business attributes, and operating hours of the business.
Review	6,990,280	Stores information about customer reviews, review ratings, review text, review dates and engagement of customers through useful, funny and cool votes.
User	1,987,897	Contains reviewer information such as review history, average ratings given, elite status, fan counts, and other profile statistics.

Each dataset has unique data identification that links it to other datasets. A review will only have one `business_id` and only one `user_id`. These relationships enable information from multiple datasets to be combined into a single integrated dataset for analysis.

The Business dataset contains the attributes of each business, the Review dataset contains the opinions and interaction of users of those businesses, the User dataset contains attributes of reviewers. Each of these sets of data only represents part of the overall picture. They can also be integrated to gain a wider picture of business performance.

2.2 Data Cleaning

In order to achieve the best possible results from the analysis, some data cleaning procedures were applied before the datasets were integrated.

Firstly, records that contains missing values in the `business_id` column were excluded from the Business dataset as the primary key used during the dataset integration will be `business_id`. Next, duplicate business records were also removed to ensure that every business was present just once. Business without category information were excluded as well since the category level analysis used business category information later.

In the Review dataset, records with no identifier for a review ere excluded, and duplicate review identifiers were deleted. Duplicate review may misrepresent review counts, average ratings and engagement data in each review, since each review should be a unique customer opinion.

Likewise, the User dataset was also cleaned by removing records with no `user_id` and duplicate user identifier. Unique user records allow for proper user information to be integrated when reviewing data.

These cleaning procedures were designed to be conservative The data was not lost, just records that would have interfered with data integrity and the ability to join were removed.

2.3 Data Integration

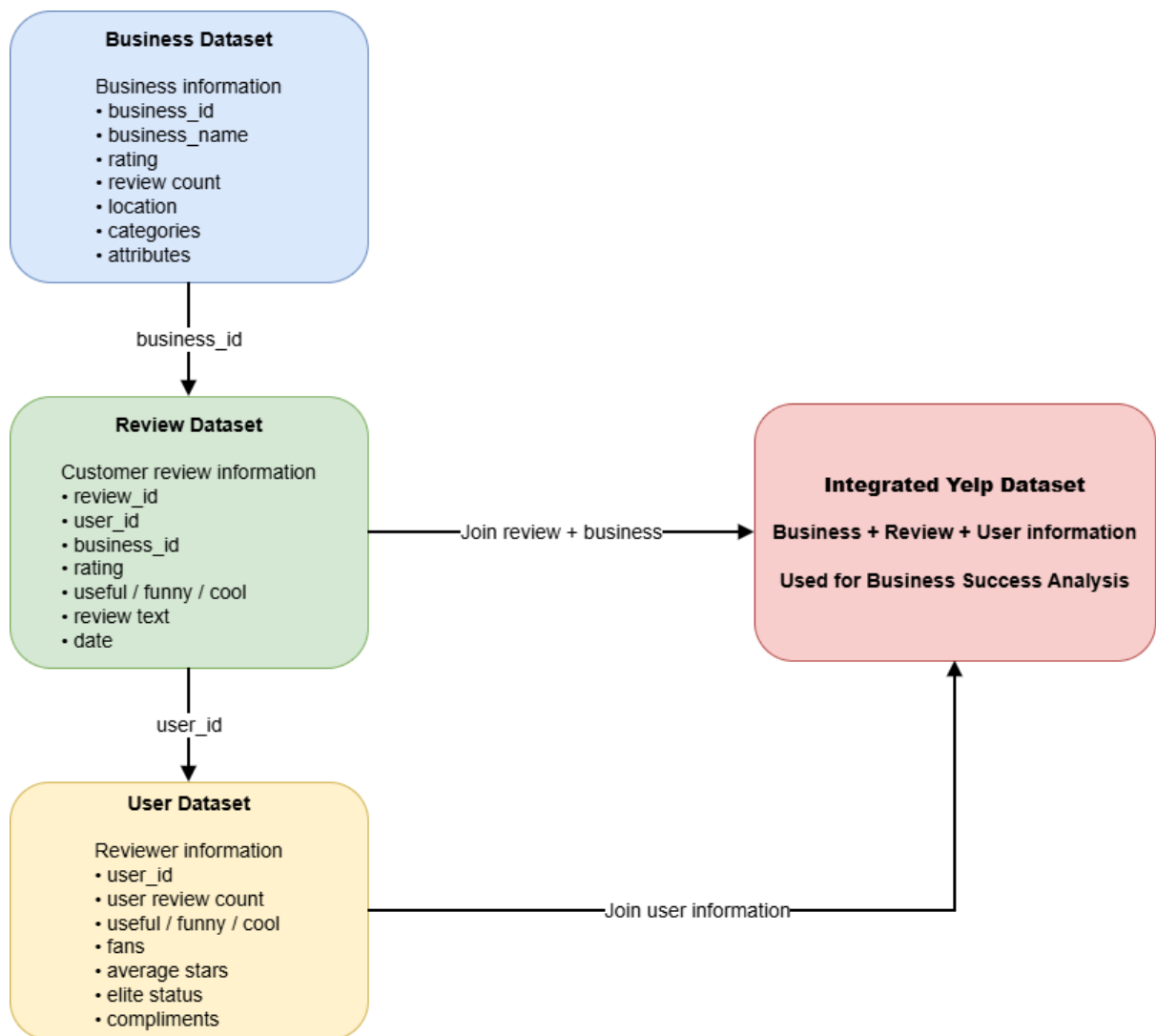
The three datasets were then joined together using Apache Spark DataFrames after performing data cleaning. There were two inner joins for the integration process.

The first step in the process was to merge the Review dataset with the Business dataset and merge them using the common field which was `business_id`. This information was amalgamated with a variety of business details like business name, location, business categories, and published Yelp ratings.

The data was then merged with the User dataset using the `user_id` column. This added the reviewer profile data which includes review history, average ratings from the users, elite status, number of fans and many more.

The integration process resulted in one integrated dataset with all the review information, business characteristics, and reviewer information in each record. This cohesive information was then fed into the algorithm called Business Success Analysis.

The workflow of the integration is shown below.



2.4 Integration Validation

Multiple validation checks were carried out after integration to confirm the correct joining of the datasets.

Validation Check	Result
Business join retained expected rows	Passed
User join retained expected rows	Passed

Final row count did not exceed original review count	Passed
Review_id remained unique after joins	Passed
Identifier columns contained no null values	Passed

Only 689 of the reviews that mentioned businesses that were not in the Business dataset were deleted by the Business join. This equates to a retention rate of 99.9901%.

The following User join deleted 33 reviews due to lack of user profiles, which gave a retention rate of 99.9995%.

In total, there were 6,990,280 reviews that were initially present, and 722 review records were deleted, which is an average of less than 1%, indicating that over 99.99% of the original data was successfully retained.

The final integrated data included:

Metric	Value
Total integrated records	6,989,558
Unique reviews	6,989,558
Unique businesses	150,243
Unique users	1,987,653

Null identifiers values	0
Duplicate review IDs	0

The results of these validations show that the integration process actually did not lose much information, and the data were kept consistent and in good referential integrity.

2.5 Purpose of the Integrated Dataset

The goal of the integration of the datasets was to build a single analytical dataset that can support the Business Success Analysis algorithm.

The integrated data enables you to calculate several performance indicators for business at once, such as:

- Average ratings of the customers' reviews
- The total number of reviews
- Customer engagement with the use of useful, funny and cool votes
- Business category analysis
- Comparison across cities and state
- Overall Business Success Scores

If there were no data integration, then these metrics would be contained in separate datasets and would not be able to be analysed together efficiently. The project used the Business, Review and User datasets as a single Spark DataFrame to create a comprehensive foundation to perform big data business analytics at scale and to show the benefits of using Apache Spark on interconnected big data.

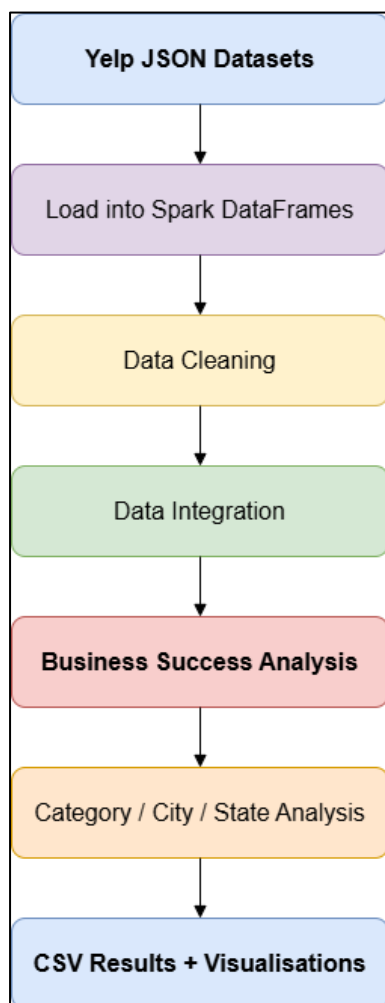
3.0 Methodology and Results: Spark

3.1 Overview of the Methodology

The goal of this research is to find successful businesses from the Open Yelp dataset using an algorithm called Business Success Analysis implemented with Apache Spark (PySpark). Apache Spark was chosen because the Review dataset is nearly seven million records which is better processed by distributed processing compared to traditional single-machine processing.

The entire process involved five steps including loading, cleaning, integrating data, analyzing the business success based on the data and visualizing the results. Each stage of the process cleaned and filtered the information for the next stage and ensured the overall result is accurate, reliable, and efficient to compute.

The figure below shows the overall Spark workflow.



3.2 Spark Implementation

The implementation was developed by using PySpark DataFrames. Spark automatically divides the big datasets and distributes the calculations over the available processing resources. Spark works on partitions of data, rather than single records, allowing it to scale much more than Hadoop does for large datasets.

Spark DataFrames were first loaded with three JSON datasets. Spark automatically deduced the data schema and translate the fields to the proper datatypes such as strings, integer, floating-point, or time-stamp during loading.

The datasets were then loaded and cleaned prior to the data being merged into one analytical dataset using an inner join. The integrated dataset was validated after which the Business Success Analysis algorithm was used.

3.3 Business Success Analysis Algorithm

The Business Success Analysis algorithm assesses businesses based on three complementary indicators of business performance, rather than just relying on customer ratings.

The indicators are shown in the table below:

Performance Indicator	Weight	Purpose
Average review rating	40%	Measures customer satisfaction
Review volume	35%	Measures business popularity

Customer engagement	25%	Captures customer interaction with Useful, Funny and Cool votes
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Measure of volume and engagement differs significantly based on the company. Some of the most well-known companies can get thousands of reviews, whereas a lot of smaller companies can get a few dozen. It would otherwise be more likely for the most popular businesses to be overwhelmingly placed in the top rankings with raw values.

This effect was minimized by taking a logarithmic transform before normalization. These metrics were then rescaled onto a common scale (0 to 1) prior to calculating the final Business Success Score.

The Business Success Score was determined using Equation (1) as shown below:

$$\text{Business Success Score} = 0.40(\text{Rating}) + 0.35(\text{Review Volume}) + 0.25(\text{Engagement})$$

In which the components denote the corresponding normalized value.

3.4 Business Eligibility Threshold

The ranking did not include all the businesses. Businesses with less than 20 matched reviews were not included in the Business Success Score calculation.

The threshold was chosen as it is believed that businesses with a few dozen or less reviews will have a less reliable average rating. For instance, a business with only two five-star ratings would seem to be doing better than a business with thousands of high ratings that does not have as much evidence of customers' ratings.

The twenty count provides sufficient reviews for determining stability and fairness of the rankings, while still providing a high volume of businesses to analyse.

Following this application of this threshold, of the 150,243 businesses in the integrated dataset, 63,487 were deemed eligible.

3.5 Implementation Workflow

The implementation of Spark consists of transformations of DataFrames and actions, including dataset loading, cleaning, the process ends with analytical output generation. The key operations used in each stage in Spark implementation is shown in the table below.

Stage	Spark Operation	Purpose
Dataset Loading	<code>spark.read.json()</code>	Load JSON datasets into Spark DataFrames
Cleaning	<code>dropDuplicates(),</code> <code>filter()</code>	Remove invalid and duplicate records
Integration	<code>join()</code>	Combines Business, Review and User datasets
Aggregation	<code>groupBy(), agg()</code>	Calculate review statistics for each business
Feature Engineering	<code>withColumn()</code>	Compute engagement metrics, logarithms and normalized values
Ranking	<code>orderBy()</code>	Rank businesses using Business Success Score

Output	<code>write.csv()</code>	Save analytical results
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Transformations were used throughout the Workflow since they are executed lazily. Spark will only do the actual computation when an action, like writing results, displaying output, occurs, thus minimizing unnecessary computation.

3.6 Validation

Several validation checks were carried out once the business metrics were computed. This is to ensure the resulting data remained consistent and suitable for further analysis. These validation checks can help examine the ratings validity, engagement values, business aggregation, identifier uniqueness and the missing values. The implementation confirmed the below:

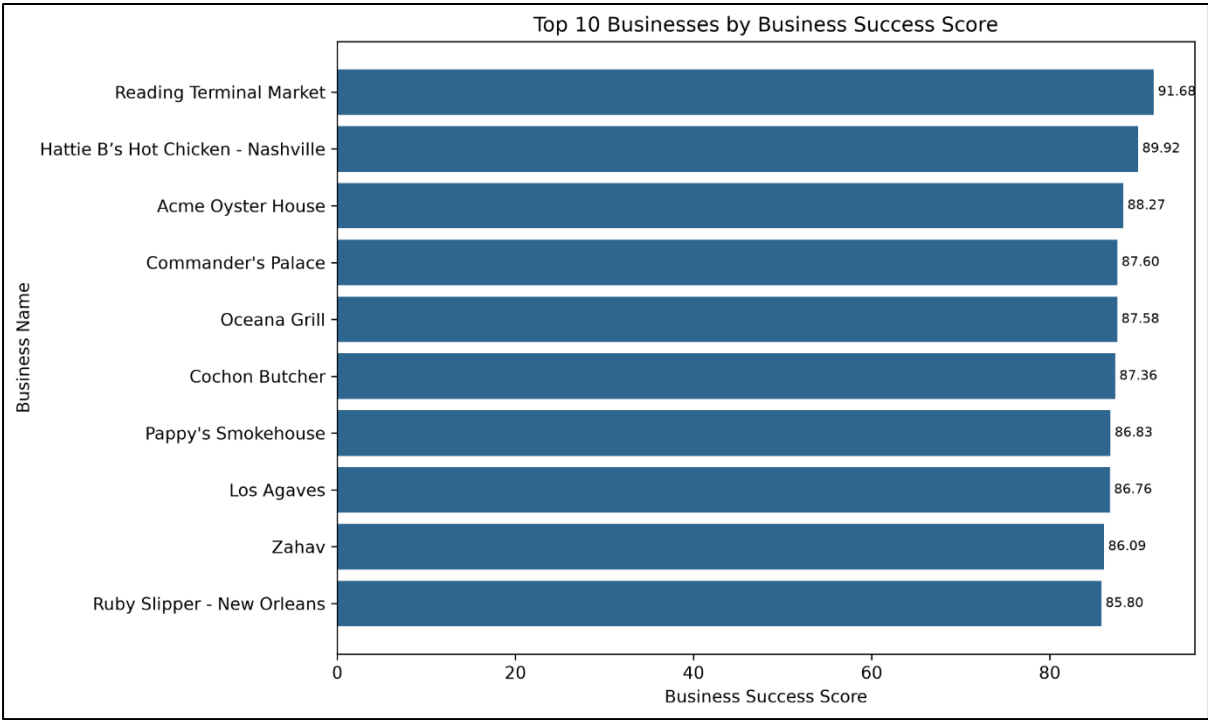
- Average review ratings were similar to valid values of 1 to 5 stars
- No business was aggregated more than once
- Tests were positives
- The values of the engagements were positive or zero
- The names of identifier columns were still unique
- There were no null identifiers following integration

The comparison was kept as a diagnostic analysis instead of an error in validation due to differences that could occur because of the updating of the data, the exclusion of some reviews, or variations in how Yelp aggregates data.

3.7 Findings

3.7.1 Top Ranked Businesses

Below are the chart and the table that represents the top 10 businesses by business success score.

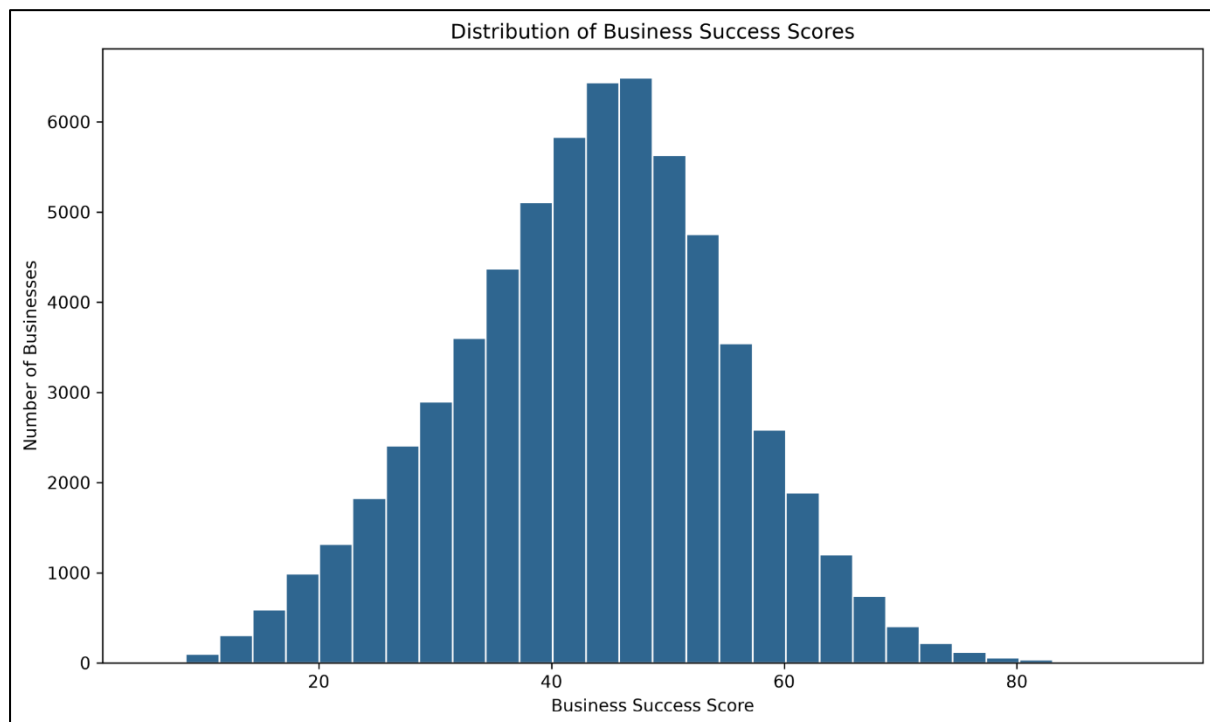


Business	City	State	Average Rating	Total Reviews	Total Engagement	Business Score
Reading Terminal Market	Philadelphia	PA	4.6054	5,778	10,609	91.6836

Business	City	State	Average Rating	Total Reviews	Total Engagement	Business Score
Hattie B's Hot Chicken - Nashville	Nashville	TN	4.4463	6,160	8,432	89.9191
Acme Oyster House	New Orleans	LA	4.1250	7,673	9,386	88.2667
Commander's Palace	New Orleans	LA	4.2920	4,969	10,371	87.6006
Oceana Grill	New Orleans	LA	4.1462	7,516	6,804	87.5822
Cochon Butcher	New Orleans	LA	4.5093	3,917	6,823	87.3553
Pappy's Smokehouse	Saint Louis	MO	4.4652	4,093	5,919	86.8325
Los Agaves	Santa Barbara	CA	4.4893	3,889	5,888	86.7582
Zahav	Philadelphia	PA	4.5487	3,173	5,766	86.0949

Business	City	State	Average Rating	Total Reviews	Total Engagement	Business Score
Ruby Slipper - New Orleans	New Orleans	LA	4.2912	5,264	4,280	85.8047

3.7.2 Interpretation of Business Success Scores



The Spark implementation was able to process 63,487 businesses that qualified. Reading Terminal Market in Philadelphia, PA, was the top ranked business. It scored 91.68 on its Business Success Score by combining:

- High number of ratings
- Over five thousand matched reviews
- More than ten thousand customer engagement votes

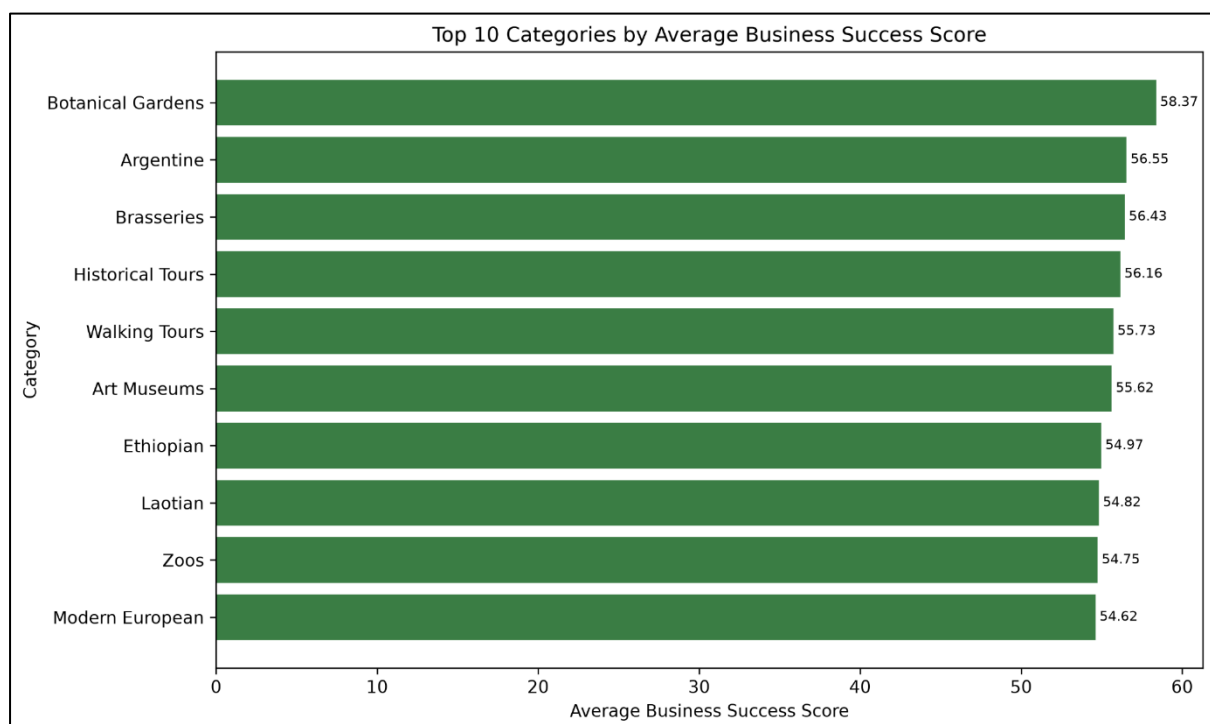
The Business Success Analysis was different from a simple ‘Star’ rating approach as it identified businesses that were successful in several areas.

Several of New Orleans’ well-known restaurants also made the top ten businesses, so it's even more important to balance customer satisfaction with maintaining popularity and customer engagement.

3.8 Category, City and State Analysis

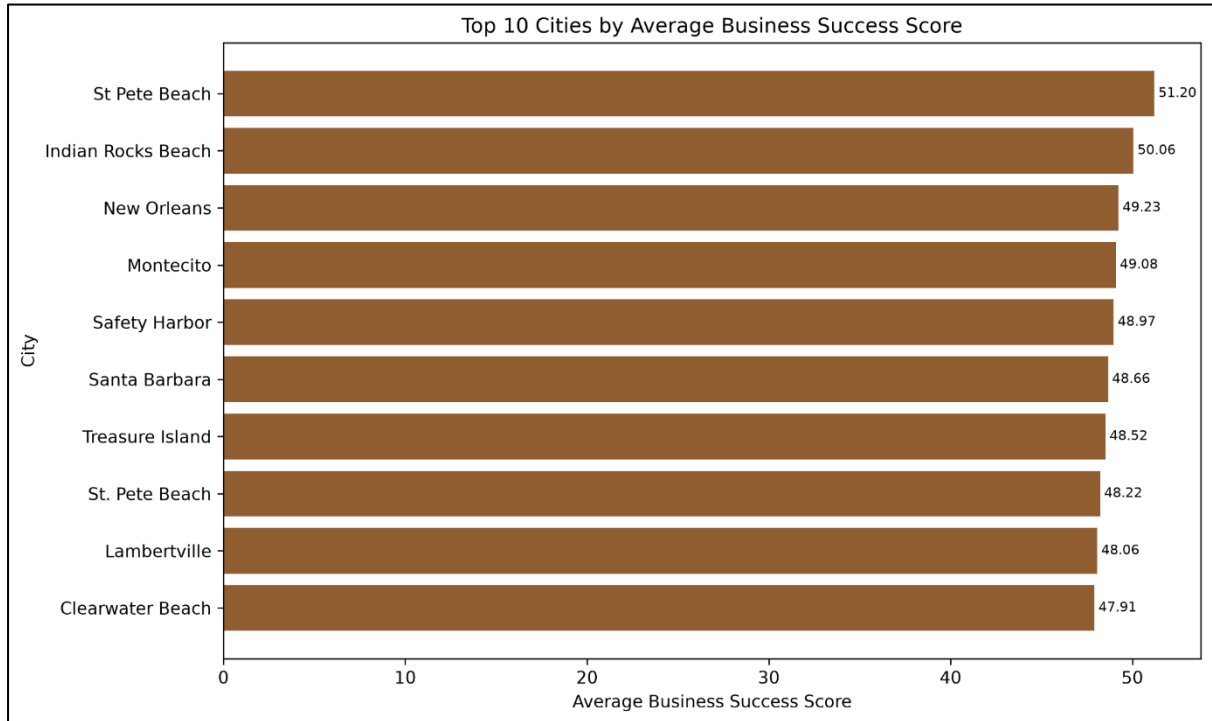
3.8.1 Category Analysis

Category level was extended by the Business Success Analysis. This was due to help identify the types of businesses that achieved the overall highest score. From the original category field, the business categories were extracted and then separated before aggregation.



Category	Businesses	Average Rating	Total Reviews	Total Engagement	Category Score
Botanical Gardens	37	4.3775	7,890	25,855	58.3730
Argentine	24	4.1355	5,507	10,334	56.5468
Brasseries	41	3.9811	16,627	28,369	56.4336
Historical Tours	84	4.5602	15,029	22,649	56.1573
Walking Tours	49	4.5361	9,239	11,196	55.7280
Art Museums	33	4.2739	5,754	20,772	55.6203
Ethiopian	44	4.1857	6,728	14,908	54.9724
Laotian	40	4.1158	7,512	11,935	54.8232
Zoos	36	3.9390	7,764	22,154	54.7537
Modern European	99	4.1170	19,696	43,192	54.6160

3.8.2 City Analysis

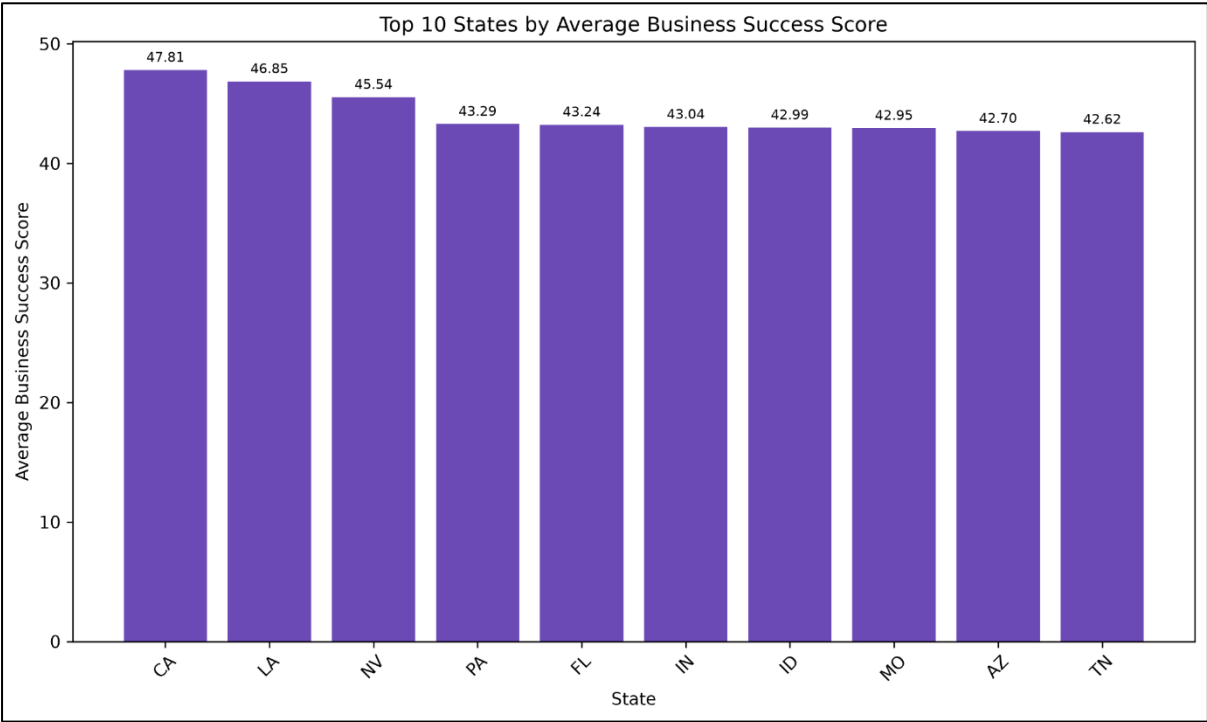


City	State	Businesses	Average Rating	Total Reviews	Total Engagement	City Score
St Pete Beach	FL	47	3.8869	10,046	10,647	51.2039
Indian Rocks Beach	FL	47	3.7910	10,286	10,449	50.0606
New Orleans	LA	3,302	3.8556	606,305	1,158,342	49.2343
Montecito	CA	43	4.0925	4,541	9,916	49.0830

City	State	Businesses	Average Rating	Total Reviews	Total Engagement	City Score
Safety Harbor	FL	61	4.0323	6,188	13,091	48.9674
Santa Barbara	CA	1,881	3.9646	250,151	536,804	48.6594
Treasure Island	FL	55	3.8172	8,928	12,187	48.5185
St. Pete Beach	FL	95	3.7570	17,535	19,608	48.2236
Lambertville	NJ	63	3.8585	6,640	12,118	48.0639
Clearwater Beach	FL	106	3.6723	21,256	29,862	47.9130

The analysis of the city shows that the highest City Score of 51.20 is achieved by St Pete Beach followed by Indian Rocks Beach and New Orleans. With substantial review volume and customer engagement, St Pete Beach have a relatively high average rating. This factor contributed to its strong overall score. Although New Orleans have performed strongly even when it has high number of businesses, the Business Success Score still considers multiple indicators of performance instead of just counting the business only. This means that cities that have strong customer satisfaction, review activity, and engagement tend to achieve higher City Scores.

3.8.3 State Analysis



State	Businesses	Average Rating	Total Reviews	Total Engagement	State Score
CA	2,605	3.9044	322,921	678,324	47.8050
LA	4,718	3.7408	710,648	1,377,352	46.8535
NV	3,686	3.6715	391,077	1,093,551	45.5350
PA	14,789	3.5839	1,412,105	3,038,269	43.2877
FL	10,796	3.6221	1,011,244	1,871,341	43.2406

State	Businesses	Average Rating	Total Reviews	Total Engagement	State Score
IN	4,822	3.6177	426,904	736,636	43.0376
ID	1,688	3.6589	130,581	242,388	42.9873
MO	4,715	3.5666	442,237	857,420	42.9464
AZ	4,263	3.5731	376,301	751,917	42.6986
TN	5,302	3.5735	548,063	847,215	42.6156

The state analysis shown by both the chart and the table indicates that California have achieved the highest State Score of 47.81. California have performed well consistently across the three performance indicators used by the Business Success Score. Besides that, the results also shows that the algorithm can be extended to broader geographical level and not just individual businesses.

3.8.4 City Threshold Analysis

Minimum Businesses per City	Cities Retained	Highest- Ranked City	Businesses Supporting Result	Average Score
10	369	Indian Shores, FL	15	51.30

20	260	St Pete Beach, FL	47	51.20
30	220	St Pete Beach, FL	47	51.20
50	166	New Orleans, LA	3,302	49.23

The Business Success Analysis was extended beyond individual businesses by aggregating results according to business categories, cities and states. Spark's transformations, split() and explode(), were used to separate the category names from the original string of categories to be used for aggregation. Minimums were also applied to the city and state analyses to prevent many of the very small city and state communities from dominating the rankings.

Four city thresholds evaluated including 10, 20, 30 and 50 qualifying businesses, were used for sensitivity analysis. The analysis indicated that 20 businesses represented the optimum number of businesses for both statistical reliability and geographical coverage. If the cutoff is raised to 30, the same top city is ranked, which shows a stable ranking.

3.9 Performance

The Business Success Analysis was completed successfully using Apache Spark.

Metric	Result
Businesses analysed	63,487

Main aggregation time	~ 99.98 seconds
Full dataset cached	No
Integrated dataset written to disk	No
Validation checks	Passed
Result files generated	CSV tables and visualizations

The implementation intentionally did not cache the full integrated dataset as previous testing suggested that there could be a possibility of Java heap memory errors if the caching was implemented. However, Spark only computes intermediate transformations when needed. This resulted in the analysis succeeding with less memory usage.

3.10 Discussion of Spark

The Yelp dataset was particularly suitable for processing with Apache Spark as the distributed DataFrame API processed millions of records for reviews efficiently, while offering succinct analytical operations like joins, aggregations and feature engineering. Spark's lazy evaluation also eliminated superfluous computation by deferring it till it was needed.

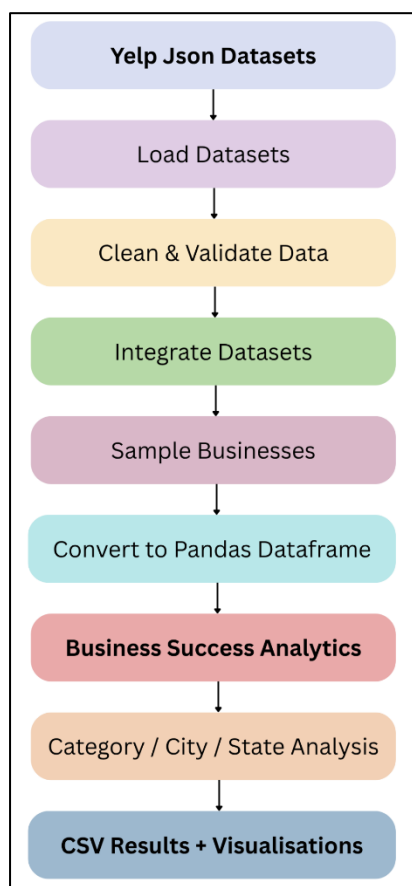
The same Business Success Analysis could be applied with other libraries like Pandas, but in very large datasets, Spark can be scaled out to multiple processing cores or multiple nodes in a cluster. This is especially suitable for applications that need to process millions of records such as big data.

4.0 Methodology & Results: Pandas

4.1 Overview of the Methodology

This research aims to identify successful businesses from the Open Yelp dataset using an algorithm called Business Success Analysis. This algorithm was implemented by Pandas to compare against the Apache Spark implementation. A key difference between the two implementations is that the Spark version processed the full dataset of 6.99 million review records while the Pandas implementation was designed to show how the memory constraints of a single computer can affect how the same analytical logic performs.

Similar to the Spark implementation design, the Pandas version also involved five main steps which includes loading and integrating data using Spark, sampling businesses to reduce the dataset size to fit within the memory limits, calculating the business success scores and creating visualisations of the results. This Pandas implementation reused a similar cleaning and integration process as the Spark version in order to ensure compatibility and fair comparison. A representative sample of businesses was taken before performing the analysis using Pandas.



4.2 Pandas Implementation

The implementation was developed by using Pandas DataFrames in VS Code to process the JSON datasets. In contrast to Spark which distributes computation across multiple cores, Pandas functions locally on a single computer with the available system memory which can cause challenges when working with large-scale data due to memory constraints.

The three datasets were first loaded into Pandas DataFrames. The datasets were cleaned by checking and handling any missing data, removing duplicates or unnecessary columns and examined to ensure that the data types were consistent and fields were correctly interpreted.

After the dataset was cleaned, they were then merged into a single dataset using inner join and the integrated datasets were validated by verifying the data consistency, missing values, structure and record counts. The cleaned and integrated Pandas DataFrame was ready for the Business Success Analysis algorithm to be implemented.

4.3 Business Success Analysis Algorithm

To ensure fair comparison with the Spark implementation, the same Business Success Analysis Algorithm was performed along with the same business performance indicators.

Performance Indicator	Weight	Purpose
Average review rating	40%	Measures customer satisfaction
Review volume	35%	Measures business popularity

Customer engagement	25%	Captures customer interaction with Useful, Funny and Cool votes
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4.4 Business Eligibility Threshold

Likewise, to the Spark implementation, businesses with less than 20 matched reviews were also excluded from the Business Success Score calculation to provide fairness and stability for the rankings.

Given that Pandas was not able to fully load the 6.99 million rows on the local machine, a business-level sampling technique was used. The fully integrated Spark DataFrame was first loaded using the same integration process as the Spark implementation. Next, a random sample of 3000 businesses was then selected using F.rand (seed=42) and all reviews for the sampled businesses were retrieved and converted to Pandas for analysis.

The details of the values can be shown in the table below:

Metric	Value
Businesses sampled	3000
Total rows converted to Pandas	131087
Unique businesses	3000
Unique reviews	131087

Businesses meeting threshold (≥ 20 reviews)	1243
Percentage of sampled businesses eligible	41.4%

This helps ensure that each selected business maintains all reviews and that the results are based on the business's complete review history.

4.5 Implementation Workflow

The key operations of the implementations were summarised in the table below:

Stage	Pandas Operation	Purpose
Dataset Loading	<code>spark.read.json()</code>	Load JSON datasets into Spark DataFrames (reused from Spark)
Cleaning	<code>Spark dropDuplicates(), filter()</code>	Remove invalid and duplicate records (reused from Spark)
Integration	<code>Spark join()</code>	Combines Business, Review and User datasets (reused from Spark)
Business Sampling	<code>Spark sample(), limit(), isin()</code>	Select 3,000 random businesses with all their reviews

Pandas Conversion	toPandas()	Convert sampled Spark DataFrame to Pandas
Aggregation	Pandas groupby(), agg()	Calculate review statistics for each business
Feature Engineering	Pandas vectorised operations	Compute engagement metrics, logarithms and normalized values
Ranking	Pandas sort_values()	Rank businesses using Business Success Score
Output	Pandas to_csv()	Save analytical results
Visualisation	Matplotlib	Generate charts for result interpretation

The integration process and conversion of the data to Pandas took up most of the processing time while the Pandas analysis was calculated swiftly. This indicates that Pandas analysis was completed very quickly as it works on a smaller sample of the data. Hence, Spark was used to handle the large-scale data, while Pandas was used for the final analysis.

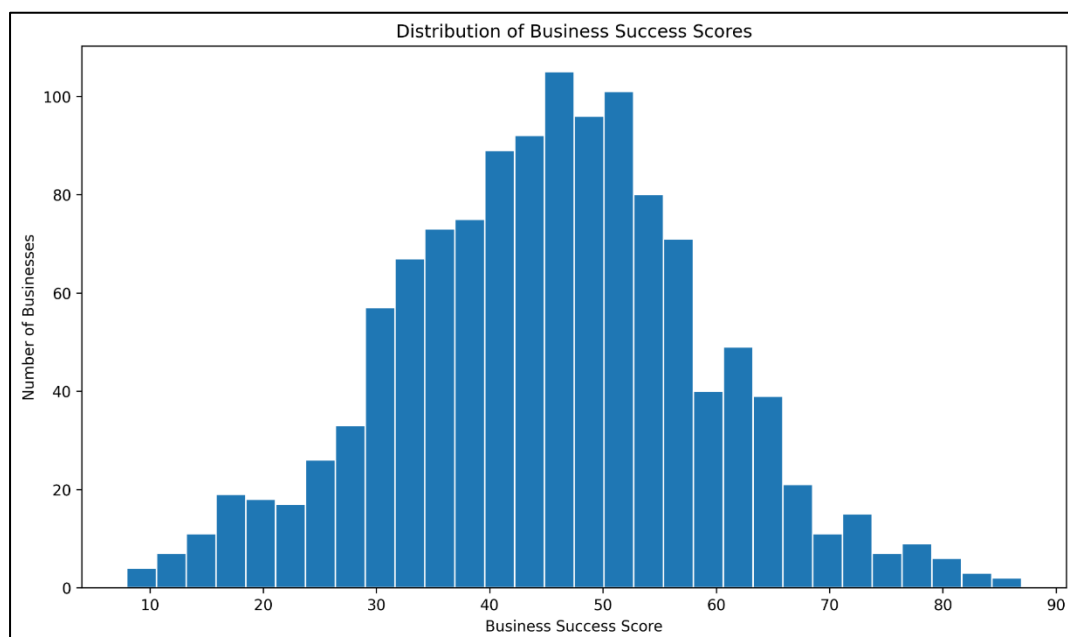
4.6 Validation

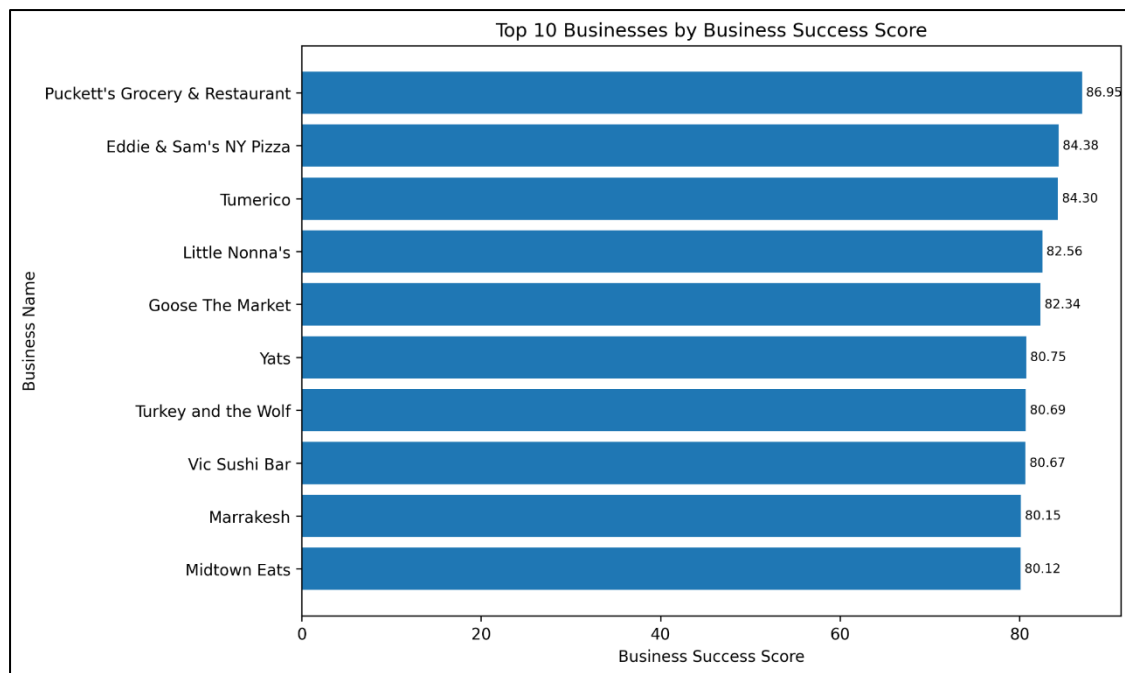
Validation checks were also performed here to ensure data integrity. The results of the validations checks can be shown in the table below.

Validation Check	Result
------------------	--------

Required columns present	Passed
DataFrame contains records	Passed
Duplicate review IDs	0
Unique businesses (sampled)	3000
Unique reviews	131087
Unique users	104096

4.7 Findings



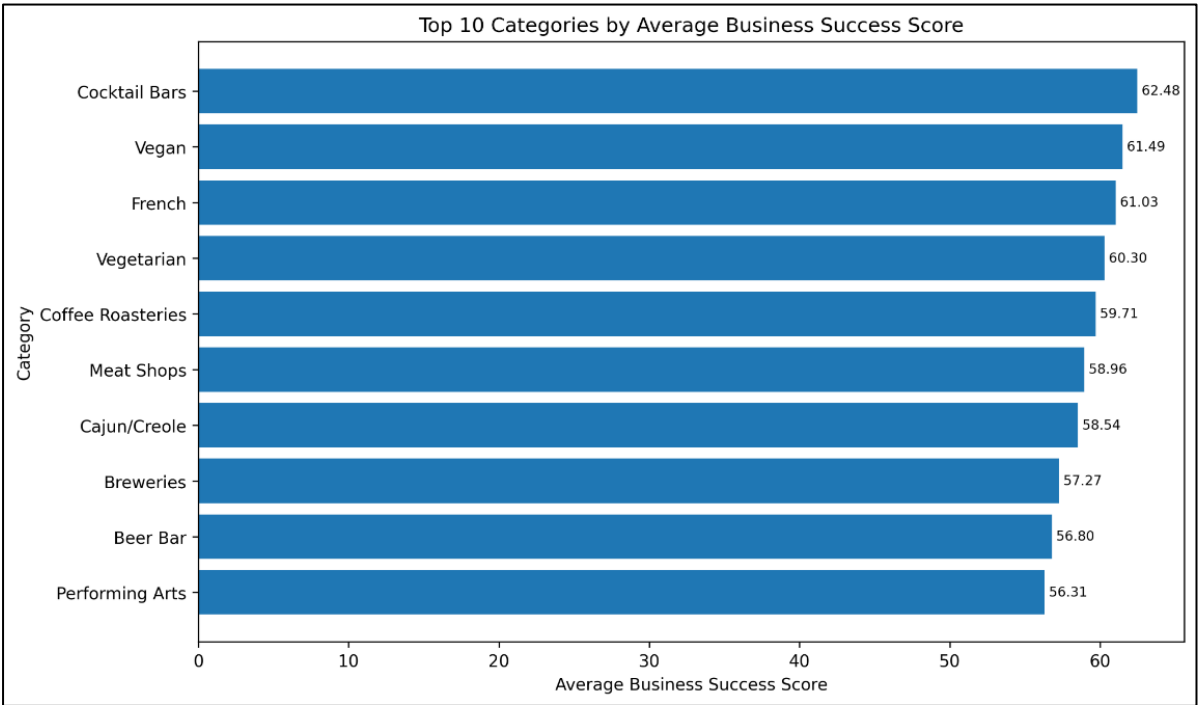


The highest ranked business result shown after calculating the Business Success Score for the qualified businesses sampled was Puckett's Grocery & Restaurant in Nashville, Tennessee with a score of 86.9548. Puckett's Grocery & Restaurant also had a high review volume of 2794, showing a normalised volume score of 1.0.

Moreover, three of the top ten businesses were located in Philadelphia and two were in Indianapolis. The majority of the top-ranking businesses were in the restaurant or food industry. The business Tumerico was ranked lower than Eddie & Sam's NY Pizza despite receiving a higher customer rating because Tumerico has substantially lower reviews and customer engagement compared to Eddie & Sam's NY Pizza. This shows that Business Score Analysis examines the overall business success instead of only relying on ratings.

4.8 Category, City and State Analysis

4.8.1 Category Analysis

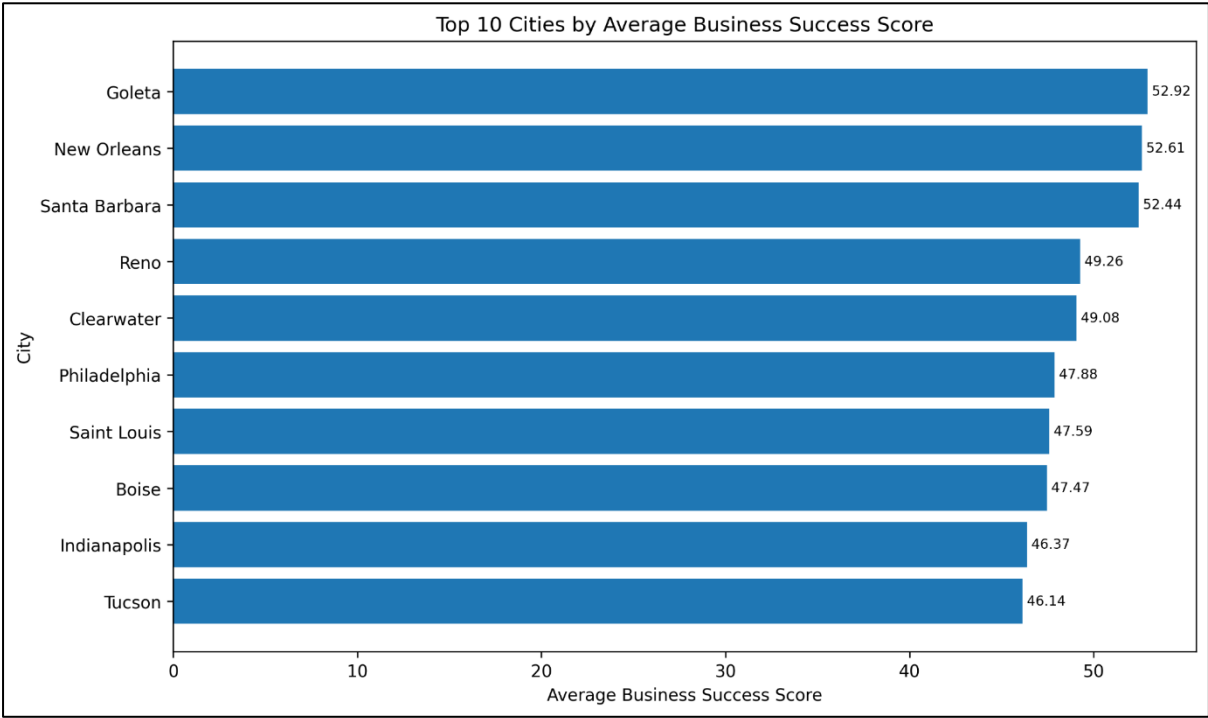


Category	Businesses	Average Rating	Total Reviews	Total Engagement	Category Score
Cocktail Bars	20	4.0000	4.0000	13,481	62.4820
Vegan	8	4.5000	1,820	3,215	61.4876
French	6	3.9167	1,659	3,013	61.0330
Vegetarian	9	4.1667	1,804	2,871	60.2956

Coffee Roasteries	5	4.4000	638	2,855	59.7061
Meat Shops	6	4.3333	927	2,586	58.9629
Cajun/ Creole	12	3.6250	4,093	5,650	58.5423
Breweries	7	4.4286	586	1,577	57.2691
Beer Bar	9	4.0556	1,503	2,167	56.8004
Performing Arts	5	4.5000	238	1,613	56.3145

Cocktail Bars ranked the highest as businesses in this category consistently achieved stronger customer ratings while also generating high review activity and customer engagement.

4.8.2 City Analysis



City	State	Businesses	Average Rating	Total Reviews	Total Engagement	City Score
Goleta	CA	11	3.9629	1,122	3,387	52.9198
New Orleans	LA	74	3.7239	11,987	21,791	52.6081
Santa Barbara	CA	29	4.0841	2,545	5,619	52.4355
Reno	NV	68	3.5316	9,350	27,558	49.2574

Clearwater	FL	22	3.7822	2,099	3,170	49.0774
Philadelphia	PA	148	3.6360	17,227	35,859	47.8785
Saint Louis	MO	45	3.4445	5,691	13,102	47.5864
Boise	ID	20	3.7451	1,391	3,418	47.4659
Indianapolis	IN	70	3.6852	5,979	12,109	46.3709
Tucson	AZ	70	3.6789	6,255	11,378	46.1439

Goleta ranked first as it has both strong customer ratings and positive engagement per business.

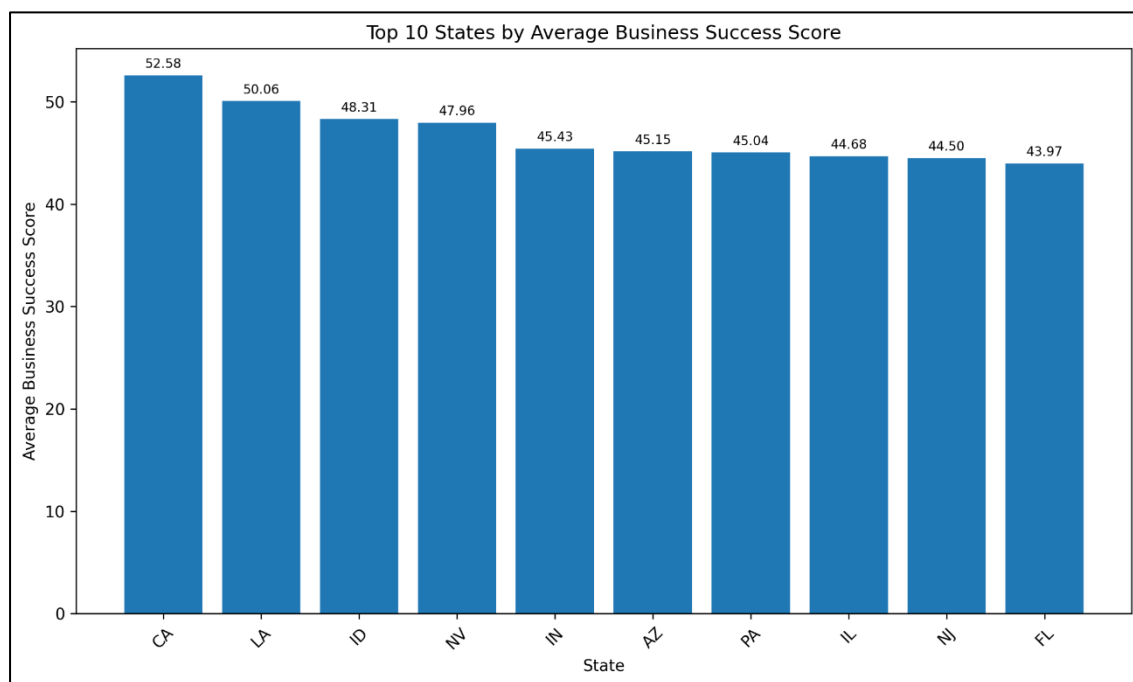
4.8.3 City Threshold Analysis

Minimum Businesses	Cities Retained	Highest-Ranked City	Businesses Supporting Result	Average Score
5	38	Goleta, CA	11	52.92
10	19	Goleta, CA	11	52.92

20	12	New Orleans, LA	74	52.61
50	7	New Orleans, LA	74	52.61

Four city thresholds were evaluated including 5,10,20 and 50 qualifying businesses. The analysis showed that the optimal threshold for both reliability in results and geographical coverage was 20 businesses.

4.8.4 State Analysis



California had the highest State Score in the sample. The state score combines the highest average customer rating among the qualifying states with high review volume and engagement.

4.9 Performance

Metric	Pandas Result
Businesses Sampled	3000
Rows in Pandas	131087
Businesses Analysed	1243
Loading, cleaning and integration	1,951.81 seconds
Business aggregation	0.24 seconds
Score calculation	0.05 seconds
Category analysis	0.03 seconds
City and sensitivity analysis	0.03 seconds
State analysis	0.01 seconds
Saving results	0.07 seconds
Total execution time	1,953.09 seconds (32.5 min)

Full integrated DataFrame cached	No
Full integrated dataset written to disk	No
Validation checks	Passed
Result file generated	CSV tables and visualisations

The integration process took up a majority of the execution time with a time of 1951.81 seconds out of 1953.09 seconds. Due to the business-level sampling, memory usage was relatively safe, resulting in the Pandas operations being completed in under 1 second. The implementation did not cache the full integrated dataset which prevented Java heap from memory errors.

4.10 Discussion of Pandas

Pandas was able to complete calculations on smaller samples of data efficiently as it completed the business, category, city and state analysis within a second on approximately 131000 rows. Additionally, Pandas has simple and readable syntax that enables functions such as filtering, aggregating and sorting to be easy to implement. In contrast to Spark, no cluster setup was necessary as it can be implemented on a local machine.

In spite of this, there were several limitations to Pandas when processing the Yelp dataset. Initially, row-based sampling of 10000 to 50000 rows was implemented but yielded insufficient businesses with 20 minimum threshold reviews. Sampling of 100000 rows results in a Java heap memory error when Spark DataFrame was converted to Pandas. Thus, business-level sampling was used to solve this issue. Hence, this shows that Pandas faces limitations in large scale data processing as the full integrated dataset could not be processed directly in Pandas on

a common local machine. This is because Pandas does not provide the distributed processing capabilities of Spark.

5.0 Comparison: Spark vs Pandas implementations

This section compares the Apache Spark and Pandas implementation of the Business Success Analysis. Although both implementations followed similar cleaning and integration processes as well as the same 20 review threshold and Business Success Score calculation, the main difference lies in the data processing approach, scalability and the memory management.

5.1 Dataset Processing Approach

Aspect	Spark (Full Dataset)	Pandas (sample)
Records processed	6,989,558	131,087
Businesses processed	150,243	3,000 (sampled)
Qualifying businesses	63,487	1,243
Percentage eligible	42.3%	41.4%

Spark implementation used the full integrated dataset with 6,989,558 review records and 63,487 businesses with at least 20 reviews. Spark was able to handle this large dataset as it distributes data across partitions and processes across multiple resources, ensuring that memory is managed efficiently through the cluster resource manager. Meanwhile, Pandas implementation only used a representative sample of the data as the test computer only has 16 GB of RAM, wrestling in inefficient and impractical processing of the full dataset. Hence, 3,000 random businesses were randomly selected, resulting in 131,087 rows and 1,243 businesses that had met the 20-review requirement. The similarity in the percentage of eligible businesses as shown in the table above (42.3% for Spark and 41.4% for Pandas) indicates that the business-level sampling created a representative subset.

5.2 Performance and Execution Time

Stage	Spark Time	Pandas Time
Loading, cleaning and integration	99.98 seconds (main aggregation only)	1,951.81 seconds
Business aggregation	Included in total	0.24 seconds
Score calculation	Included in total	0.05 seconds
Category / City / State analysis	Included in total	0.07 seconds
Saving results	Included in total	0.07 seconds
Total execution time	99.98 seconds	1953.09 seconds

As shown in the table above, the Spark implementation completed the full process in 99.98 seconds. This includes loading, cleaning, integration, analysing, calculation and producing outputs. Hence, showing that Spark was able to handle the full dataset efficiently. Spark was able to process all 6.99 million records without exceeding the memory limit, indicating its ability to scale datasets that are otherwise too large for Pandas to process efficiently.

In contrast, the Pandas implementation took up 1953 seconds which is approximately 32 minutes to complete the full analysis. However, 1951.81 seconds which is 99.9% of the time was spent on Spark integration stage to load, clean and combine the datasets before sample was

created. The aggregation, scoring, analysis and saving the outputs took up approximately 0.43 seconds, indicating that Pandas was able to complete analysis efficiently when working with data that is within the memory limits but preparing the data for Pandas was time intensive.

5.3 Business Success Analysis Results

Aspect	Spark (Full Dataset)	Pandas (sample)
Highest-ranked business	Reading Terminal Market	Puckett's Grocery & Restaurant
Highest-ranked category	Botanical Gardens	Cocktail Bars
Highest-ranked city	St Pete Beach, FL	Goleta, CA
Highest-ranked state	California	California

A common result of both implementations is that California ranked the highest state which shows that the sampling methods were able to maintain the general pattern of the data, producing consistent results at the broader level. On the other hand, the results for highest ranked business, category and city differ between the implementations as Pandas had only analysed a sample of the full dataset instead of the complete dataset, producing different rankings. Spark identified Reading Terminal Market as the highest ranked business while Pandas ranked Puckett's Grocery & Restaurant the highest.

6.0 Discussion: Big Data advantages and interpretation

6.1 Advantages of Big Data Processing

The main advantages of using big data processing in contrast to conventional methods like Pandas was the ability to analyse large volumes of data without exceeding the memory limits (Gusman et al., 2022). In contrast to Pandas, Spark was able to process the full 6,989,558 records without encountering memory constraints. Several features of big data processing made this possible including lazy evaluations, partitioning, memory management as well as no full DataFrame caching.

6.1.1 Lazy Evaluations

Lazy evaluations refer to how transformations such as filtering or grouping are not computed when they are written but are only executed once an action was performed. This enables Spark to compose a physical execution plan, allowing operations to be organised into stages and processed efficiently as data locality could be maintained (Capuccini et al., 2020).

6.1.2 Data Partitioning

Apache Spark uses Resilient Distributed Datasets (RDD) to store the data, dividing the data into logical partitions to be processed on different Spark cluster nodes in parallel (Singh et al., 2023). As shown in this assignment, partitioning allowed Spark to distribute the large datasets into smaller and more manageable portions so that the full dataset does not need to be handled on a single machine's memory.

6.1.3 Memory Management

Another advantage to Spark is its ability to manage memory across the cloud resources. Memory usage varies across tasks such as computing, shuffling or storing data. According to Myung and Choi (2021), the required memory usage is dependent on the workload and size of the input data and insufficient memory can result in longer processing time or errors in job completion. Hence, memory availability has a crucial effect on Spark performance and should be managed efficiently especially when the dataset size is large.

6.1.4 No Full DataFrame caching

In this project, the implementation did not cache the full integrated DataFrame. This means that Spark kept the DataFrame in memory so that no re-computation was required. Given that caching the large dataset would be memory intensive, this would decrease the memory usage and pressure.

6.2 Spark and Pandas: Interpretation of the Results

Spark was able to fully process the integrated Yelp dataset while Pandas had to turn to sampling the dataset due to memory constraints. This resulted in several differences in outcomes for the business success analysis rankings between the Spark and Pandas implementation. The Spark results were able to provide a complete view of the datasets as all eligible businesses that passed the review thresholds were included while the Pandas results only provided the sampled businesses. Hence, this reinforced the statement by Gusman et al. (2020), which highlighted that big data processing can help improve decision making for organisations as it is able to analyse large datasets with more accuracy, gaining a more comprehensive insight that can help optimise their performance.

7.0 Conclusion: Overall findings and recommendations

7.1 Overall Findings

This paper successfully shows the difference between big data processing techniques and conventional data processing methods when applied to analyse business performance using the Yelp Open Dataset. The Business Success Analysis algorithm was implemented by using both Apache Spark and Pandas and followed the same data preparation process, thresholds and Business Success Score indicators and calculations. Spark was able to process the complete integrated dataset of 6989558 review records and identified 63487 qualifying businesses, while Pandas analysed a representative business sample of 131087 records containing 1243 qualifying businesses.

The Spark implementation completed the full analysis in 99.98 seconds, identifying Reading Terminal Market in Philadelphia as the highest ranked business with a score of 91.6836 while California was the highest ranked state. In contrast, the Pandas version completed the full process in 1953 seconds and found Puckett's Grocery & Restaurant in Nashville to be the highest ranked business with a score of 86.9548. Similar to the Spark implementation, California was also ranked the highest state with high average rating and engagement.

By comparing the execution times of the Spark and Pandas implementation, the paper showed the clear distinction in strengths. Spark performed significantly better for processing the complete Yelp dataset due to its distributed processing, lazy evaluation, memory management and zero DataFrame caching. On the other hand, Pandas completed the actual analysis quickly once the sampled data was loaded into the memory, completing calculations in less than a second. However, Pandas was unable to process the full integrated dataset, relying on Spark to prepare and integrate the dataset as well as business sampling to select a representative sample. Thus, the findings demonstrate that execution time, dataset size and processing tasks are important indicators of each framework's performance.

7.2 Limitations

The paper has several limitations such as geographical limitations as the Yelp Open dataset mainly covered businesses in the United States. Another constraint was that the review text was not included in the Business Success Score, resulting in potential useful opinions or customer sentiment being omitted. Thirdly, the business success score formula was selected based on the analytical design of the study which may produce different rankings if the weightage of each indicator were altered. Moreover, the Pandas implementation relied on only a sample of 3000 businesses, which may result in different outcomes compared to full dataset analysis.

7.3 Recommendations

Future works can consider including review text or sentiment analysis to enable a more comprehensive view of the customer satisfaction. Moreover, multiple sampling runs can also be performed with different random seed values to evaluate the Randas results while Spark can also be deployed on a cloud-based cluster for improved scalability and to test how well the implementation is able to perform under larger big data environments.

7.4 Final Conclusion

In conclusion, this paper successfully compared the distinct strengths and performances of Spark and Pandas in big data analysis. The overall findings suggested that Spark is more suitable for large-scale big data processing while Pandas was appropriate for smaller datasets. Hence, framework or tools usage should depend on the dataset size, available computing resources as well as the computational requirements. Big data processing framework can help to support organisations in analysing large volume of customer review data, identifying trends and insights for enhanced business decision making for better performance.

Individual reflection

Ayu Wen Li (22017867)

My main contribution in this assignment was developing the Pandas implementation of the Business Success Analysis and comparing the results between the Spark and Pandas implementations. Hence, this task allowed me to gain practical experience in using Pandas to load, manipulate and analyse the integrated data. Through these tasks, my knowledge and experience in handling DataFrame operations such as transformation, grouping and filtering had improved.

A key understanding I learned from this assignment was the difference in handling large datasets between Pandas and Spark. Initially, I thought that the Pandas implementation would be able to process the full large volume of records albeit significantly slower compared to the Spark version. However, I repeatedly encountered memory errors when converting 1,000,000 rows to Pandas. This showed me the practical limitations of single-machine processing and provided me a better understanding of why big data processing technologies such as Spark are essential for big data applications.

Moreover, I also learned that the data preparation stage can have a high impact on the execution runtime while comparing the Pandas and Spark performance as most of the execution time for the Pandas implementation was spent on preparing and integrating the full dataset. In dealing with the long execution time for the Pandas workflow, I learned to manage it by conducting smaller tests and validation checks during the development stage for higher efficiency and problem identification. In conclusion, I gained a clearer understanding of how Pandas is more suitable for smaller datasets for quick analysis, readability and prototyping while distributed frameworks are more appropriate when working with big data that exceeds the memory limits of a single computer.

(274 words)

Keertana A/P Subramaniam (23109614)

My primary contribution to this study was to develop Spark implementation of the Business Success Analysis. This includes loading and preprocessing the JSON datasets from Yelp, integrating all three datasets which were the Business, Review and User dataset by using Spark joins, validating outputs and recording the execution time. I also generated visualizations for the Spark outputs, and wrote the report by covering the Introduction, Dataset Description and Integration, Methodology, Implementation and Results for Spark sections. Through these tasks, I have gained practical experience when it comes to using PySpark DataFrames especially to process large-scale data that involves cleaning, joining, aggregating, and transforming interconnected datasets.

One of the key lessons I learnt was how large datasets like Yelp dataset can be handled by Apache Spark effectively through distributed processing and lazy evaluation. By working with this dataset that contains almost seven million review records made me realize that joins and aggregations operations can be very demanding computationally. Additionally, the memory usage and execution can be affected by decisions such as whether to cache a large DataFrame. Besides that, I also understood that validating the data after integration by checking the counts of the record, duplicate identifiers, and null values is very important to ensure the results remain reliable for analysis.

By developing the Business Success Analysis, it has improved my understanding on feature engineering and data aggregation. To provide a broader measure of business performance instead of just relying on these star ratings, I have learnt how weighted and normalized metrics can be used to combine ratings of the review, review volume, and customer engagement. Furthermore, by creating the category, city, and state analysis, it has helped understand how similar analytical approaches can be extended to identify broader patterns in the dataset.

Overall, this study has strengthened my ability to implement a Spark-based data analytics workflow that is complete including data preparation and integration, analysis, validation, performance evaluation and visualization. This study has also helped me gain a better understanding of why distributed processing is very useful especially when working with datasets that are large-scale.

(348 words)

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Appendix

GitHub Link: <https://github.com/Keertana0307/Yelp-BigData-Analytics>

Official Dataset Source: <https://business.yelp.com/data/resources/open-dataset/>